# Cracking the Black Box

## Portfolio Replica Strategy — BC3 final deliverable

PoliMI Fintech — Business Case 3

# Problem and data

**Goal.** Replicate a hidden "Monster Index" using only liquid Bloomberg futures, subject to a regulatory VaR cap.

$$r_t^{\text{target}} = 0.50\, r_t^{\text{HFRX}} + 0.25\, r_t^{\text{MSCI World}} + 0.25\, r_t^{\text{Global Bond}}$$

**Universe (11 futures).**

- Rates: RX1, TY1, DU1, TU2
- Equity: ES1, NQ1, VG1, TP1, LLL1
- Commodity: GC1, CO1

**Constraints.**

- Sample: weekly, Oct 2007 – Apr 2021 ($\approx$ 705 obs).
- Gaussian VaR cap: $-\Phi^{-1}(0.01) \cdot \hat{\sigma}_w \cdot \sqrt{4} \leq 8\%$.
- Gross-exposure cap: $\sum_j |w_j| \leq L$.
- Transaction cost: $\tau = 5$ bps on L1 turnover.

# Notebook structure — seven Parts

**Methodological arc, not chronological.**

| Part | Content |
|---:|---|
| I | Setup, data loading, EDA on $(X, y)$ |
| II | Backtest harness, transaction-cost model, VaR/GE/IR/TE definitions |
| III | Ridge control baseline + rebalancing cadence × leverage sweep |
| IV | Linear benchmark family via predict-then-optimize (OLS / Ridge / Lasso / EN / Huber) |
| V | Kalman filter / state-space (static, EM, regime-switching) |
| VI | Deep learning weight generator (MLP) |
| VII | Consolidated comparison + recommended configuration |

**Data flow.**

`prices` → $(X, y)$ panel → `harness.run_rolling_backtest` → {Ridge, Linear PO, Kalman, NN} → `results/*.pkl` → master table.

# Part II — backtest methodology

**Rolling walk-forward.** At every rebalance week:

- Fit on the trailing 104-week window.
- Predict $w_t$; hold weights until the next rebalance.
- No look-ahead — the training slice ends strictly before evaluation.

**Contract metrics.**

$$\mathrm{TE} = \sqrt{52\,\mathrm{Var}(r_t^{\mathsf{rep}} - y_t)}, \qquad \mathrm{IR} = \frac{52\,\mathbb{E}[r_t^{\mathsf{rep}} - y_t]}{\mathrm{TE}}, \qquad \bar{T} = \mathbb{E}_t\Big[\textstyle\sum_j |\Delta w_{t,j}|\Big]$$

**Transaction-cost subtraction.** Costs hit on rebalance dates only:

$$c_t = \tau \cdot \sum_{j=1}^{n} |w_{t,j} - w_{t-1,j}|, \qquad \mathrm{net\_IR} = \frac{52\,\mathbb{E}[r_t^{\mathsf{rep}} - c_t - y_t]}{\mathrm{net\_TE}}$$

**Risk projection layer.** Gross-exposure cap first ($\sum |w_j| \leq L$), then iterative historical-VaR shrinkage until $\mathrm{VaR}(w_t) \leq 8\%$.

# Part III — Ridge control + constraint sweep

**Canonical control baseline.** Ridge with $\alpha = 0.01$, $\Delta t = 4$ weeks. Frictionless vs. net-of-cost comparison fixes the headline cost drag.

**Cadence $\times$ leverage sweep.** Six base configurations plus a 6-axis Cartesian grid:

$$\Delta t \in \{1, 2, 4, 8, 12\}, \quad L \in \{1.0, 1.5, 2.0\}, \quad \alpha \in \{10^{-4}, 10^{-3}, 10^{-2}\}, \quad \rho_{\mathsf{EN}} \in \{0.2, 0.5, 0.8\}$$

**Constraint layer for every configuration:**

- Fit MinMax-scaled ElasticNet on the trailing 104-week panel.
- Project: $w \mapsto \Pi_{\mathcal{W}}(w)$ with $\mathcal{W} = \{w : \sum_j |w_j| \le L\}$.
- Shrink iteratively until $\mathrm{VaR}_{\mathsf{hist}}(w) \le 8\%$.

**Composite ranking.** Z-score $\{\mathrm{IR}, \mathrm{net\_IR}, \rho, -\mathrm{TE}, -\bar{T}, -\mathrm{MDD}\}$, average, sort.

# Part IV — linear benchmark family

**Predict-then-optimize.** Each rebalance:

1. **Alpha.** Fit a classical linear model on $(X_{\text{tr}}, y_{\text{tr}})$; extract $\mu = \mathrm{coef\_}$.
2. **Risk.** Ledoit–Wolf shrunken sample covariance $\Sigma$ (annualised).
3. **Optimize.** Convex problem (CVXPY):

$$\max_{w} \ \mu^\top w - \tfrac{1}{2}\lambda\, w^\top \Sigma w - \tau\, \|w - w_{t-1}\|_1 \quad \text{s.t.} \quad \|w\|_1 \le L, \ \ |w_j| \le 0.5$$

**Leaderboard sweep.**

$$\{\text{OLS, Ridge}_\alpha,\ \text{Lasso}_\alpha,\ \text{ElasticNet}_{(\alpha,\rho)},\ \text{Huber}_\delta\} \times \{\text{all-11, top-5}\}$$

**Top-$K$ pre-selection.** At each rebalance, keep the $K$ futures with highest $|\mathrm{corr}(r_j, y)|$ *computed inside the training window* — no leakage.

**Sanity check.** With $\lambda = 2$, slack constraints, and no Ledoit–Wolf shrinkage, the convex layer recovers the OLS fit to within solver tolerance.

# Part V — Kalman filter / state-space

**State-space model.** Random-walk weights, scalar return as observation:

$$\begin{aligned} x_t &= x_{t-1} + Bu_t, & u_t &\sim \mathcal{N}(0, I), & B &= \sigma_w I \\ y_t &= r_t^\top x_t + D\varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, 1), & D &= \hat{\sigma}_y \end{aligned}$$

**Three variants compared.**

- **Static $\sigma_w$ grid.** $\sigma_w \in \{10^{-4}, 10^{-3}, 10^{-2}\}$; warm-started from Ridge on the first 52 weeks.
- **EM-tuned noise.** `pykalman.KalmanFilter.em()` fits $(B, D)$ jointly, capped at 3 iterations (the 1-obs / 11-state system is under-identified at higher $n$).
- **Regime-switching.** 2-state Gaussian HMM on rolling realised target volatility selects $\sigma_w \in \{\sigma_w^{\text{calm}}, \sigma_w^{\text{stressed}}\}$.

**Leakage discipline.** The HMM is fit on the 104-week warm-up window only and then *frozen*; subsequent regime labels are produced by `.predict`, not refit. Without this, the HMM sees post-train data.

**Risk projection.** Same Gaussian VaR cap $\mathrm{VaR}(\hat{x}_t) \leq 8\%$ applied after every filter update.

# Part VI — deep learning weight generator

**End-to-end framing.** The network outputs the *weights*, not the returns:

$$w_t = f_\theta(\phi_t), \qquad r_t^{\text{rep}} = w_t^\top r_t.$$

**Features** $\phi_t$ (lagged by one step via `shift(1)` — no contemporaneous information):

- Trailing compounded returns at $k \in \{4, 12, 52\}$ weeks per asset.
- Trailing realised volatility at $k \in \{12, 52\}$ weeks.
- Regime indicators: 12-week MSCI World return, 12-week bond trend, 12-week target volatility.
- Warm start: previous Ridge weight from a 52-week trailing fit ($\alpha = 10^{-3}$).

**Architecture.** `WeightMLP`: $\phi_t \in \mathbb{R}^{d_\phi} \to 64 \to 32 \to w_t \in \mathbb{R}^{11}$, ReLU + dropout 0.1, optional gross-exposure projection inside the forward pass.

# Part VI — loss, training, rolling backtest

**Loss — forward-window tracking-error variance + turnover penalty:**

$$\mathcal{L}(\theta) = \mathrm{Var}_{s\in[t,\,t+H-1]}\big(w_t^\top r_s - y_s\big) + \lambda\,\big\|\Delta w_t\big\|_1, \qquad H = 12,\ \ \lambda \in \{0, 10^{-3}, 10^{-2}\}$$

**Training regime.**

- Chronological 60 / 20 / 20 split — no shuffling across the boundary.
- Adam, lr $10^{-3}$, weight decay $10^{-5}$, batch 64, max 500 epochs, patience 30.
- Early stopping on annualised validation TE.

**Rolling out-of-sample backtest.** `run_nn_rolling_backtest` retrains on the trailing 156-week window at each rebalance $\Delta t \in \{1, 4\}$, predicts $w_t$, applies the Gaussian VaR projection, and warm-starts the next fit from the previous state dict.

**Why variance, not MSE?** A constant tracking bias washes out under VaR rescaling; the variance directly targets the IR denominator.

# Part VII — consolidated comparison

**Synthesis.** Each track persists its result as a pickle; Part VII loads them all and ranks by net IR:

$$\{\text{Ridge control, Linear PO}_*,\ \text{Kalman}_*,\ \text{Kalman EM, Kalman regime, NN}_{\Delta t=1},\ \text{NN}_{\Delta t=4}\}$$

**Reported columns:**

$$\{\mathrm{IR},\ \mathrm{net\_IR},\ \mathrm{TE},\ \rho,\ \overline{\mathrm{GE}},\ \bar{T},\ \mathrm{VaR},\ \mathrm{cost\_drag},\ \mathrm{MDD}\}$$

**What the table answers.**

- **Best IR / best $\rho$.** Joint signal selection plus rebalance discipline.
- **Best net IR.** Captures the cost penalty for high-turnover models.
- **Crisis survival (2008-09, 2020-02→04).** Configurations that keep TE bounded and avoid VaR breaches when tails go fat.

**Spotlight plots.** Cumulative return vs. target, rolling 26-week correlation, weights heatmap over time for the winning configuration.

# Reading the leaderboard

**What the analysis shows.**

- **Cadence sets the cost/fit trade-off.** Weekly rebalancing maximises tracking quality but pays the heaviest 5 bps drag; quarterly rebalancing minimises drawdown but loses tracking precision. The constraint sweep makes the trade-off explicit.
- **Long rolling window beats recent adaptation.** The 3-year window dominates 1-year on net IR for every linear model — a stable signal beats reactive fitting on weekly data.
- **Predict-then-optimize closes the constraint gap.** The convex layer applies the $L_1$-gross and per-asset $|w_j|$ caps inside the optimisation; Ridge with post-hoc rescaling pays a small TE premium for the same risk profile.
- **Kalman + EM matches the static $\sigma_w = 10^{-3}$ grid point.** EM is a useful sanity check; the regime-switching variant narrows TE in 2008 at the cost of more turnover.
- **The MLP rivals the linear leaderboard** when warm-started from Ridge and penalised on turnover; without $\lambda\|\Delta w_t\|_1$, NN variants over-trade and lose net IR.

# Recommended configuration

**Deployable choice (composite-score winner).**

- **Model:** PO + ElasticNet on top-5 futures (pre-selected by training-window correlation).
- **Cadence:** $\Delta t = 4$ weeks — monthly rebalancing balances tracking and turnover.
- **Risk:** gross-exposure cap $L = 1.5$, iterative VaR cap $\leq 8\%$.
- **Transaction cost:** 5 bps on L1 turnover; budget $\tau \in \{2, 5, 10\}$ bps stress-tested.

**Why this configuration.**

- **Net IR survives the cost sweep.** Monthly cadence keeps $\bar{T}$ low; top-5 prunes redundant exposures.
- **Crisis robustness.** Composite ranking weights MDD and TE alongside IR; the winner is consistent across the 2008 and 2020 windows.
- **Interpretability.** The linear coefficients are portfolio weights — the strategy is transparent for a regulatory audit.

## Takeaways

- **Methodology before models.** The Part II harness fixes rolling-window discipline, transaction-cost accounting, and the VaR / GE projection layer *once*; every Part III–VI model plugs into the same evaluation surface.
- **Leakage discipline runs end-to-end.** Top-$K$ pre-selection inside the training window, HMM frozen on the warm-up slice, train-fold standardisation for the NN, `shift(1)` on every lagged feature — every fit-stage is closed against future data.
- **Turnover discipline is non-optional.** Without explicit penalisation, every adaptive model (NN, regime-switching Kalman) over-trades and loses on net IR.
- **Constraint sweep is the deployment lens.** Cadence and leverage choices dominate model choice once costs and risk caps are in play; the composite score makes the trade-off auditable.
- **Single pickle contract.** `ReplicaResult` flows through every track unchanged — that is what lets Part VII be a one-page comparison rather than five fragmented narratives.

**Thank you — questions?**